Database

ThS. Nghi Hoàng Khoa - khoanh@uit.edu.vn

Relational vs Non – Relational

	Relational Database	**Non-Relational Database**
Data Integrity	Enforces data integrity through relationships and constraints	Flexible schema allows for easy data updates
Structured Queries	Supports complex queries using SQL	May offer simpler, faster query performance
Scalability	May not scale as well for large datasets	Highly scalable and can handle big data and high traffic
ACID Transactions	Provides ACID (Atomicity, Consistency, Isolation, Durability) transactions	May sacrifice ACID properties for improved performance or scalability
Consistency	Ensures strong data consistency	May offer eventual consistency or tunable consistency
Schema Evolution	Schema changes can be complex and time-consuming	Allows for easy schema changes and flexibility in data model
Examples	MySQL, PostgreSQL, Oracle	MongoDB, Cassandra, Redis, DynamoDB

Relational Database

This lecture assume you already pass IT004 and have knowledge of SQL Queries

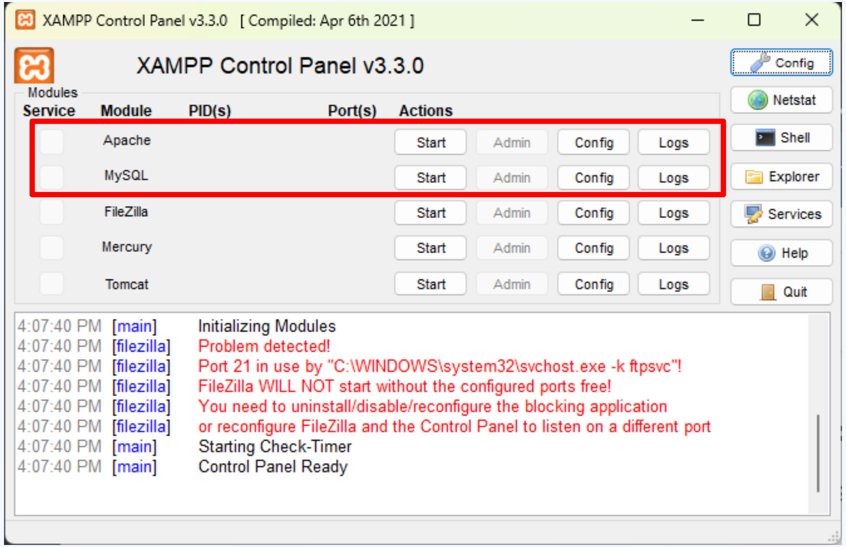
See SQL Tutorial (w3schools.com) if you need revision

MySQL

This course will use MySQL as Database Server and PHPMyadmin as DBMS

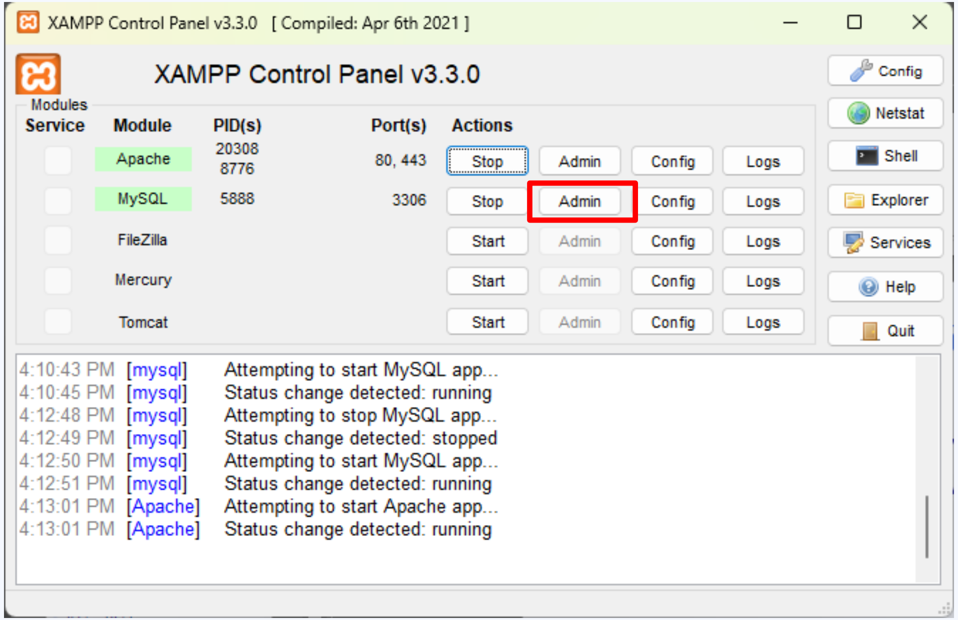
Install Xampp: XAMPP Installers and Downloads for Apache Friends

Start both PHP and MySQL Server and you're ready to go

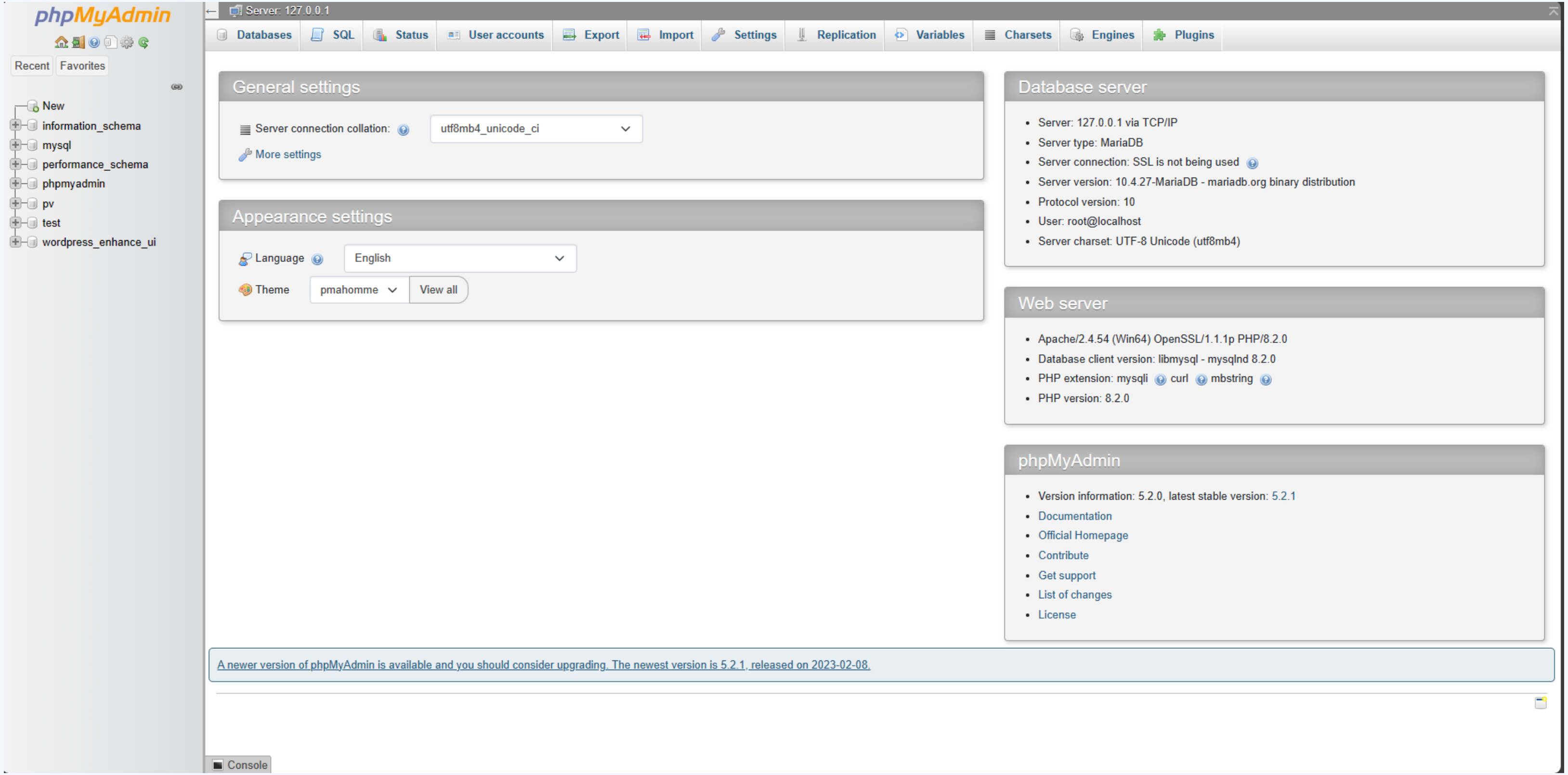


MySQL

Click on Admin to open phpMyAdmin DBMS site on your local machine

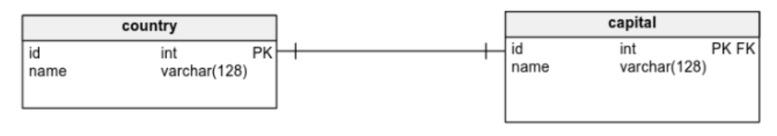


MySQL


phpMyAdmin
Recent Favorites
New
information_schema
mysql
performance_schema
phpmyadmin
pv
test
wordpress_enhance_ui
Server: 127.0.0.1
Databases SQL Status User accounts Export Import Settings Replication Variables Charsets Engines Plugins
General settings
Server connection collation: utf8mb4_unicode_ci
More settings
Appearance settings
Language English
Theme pmahomme View all
Database server
Server: 127.0.0.1 via TCP/IP
Server type: MariaDB
Server connection: SSL is not being used
Server version: 10.4.27-MariaDB - mariadb.org binary distribution
Protocol version: 10
User: root@localhost
Server charset: UTF-8 Unicode (utf8mb4)
Web server
Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.2.0
Database client version: libmysql - mysqlnd 8.2.0
PHP extension: mysqli curl mbstring
PHP version: 8.2.0
phpMyAdmin
Version information: 5.2.0, latest stable version: 5.2.1
Documentation
Official Homepage
Contribute
Get support
List of changes
License
A newer version of phpMyAdmin is available and you should consider upgrading. The newest version is 5.2.1, released on 2023-02-08.
Console

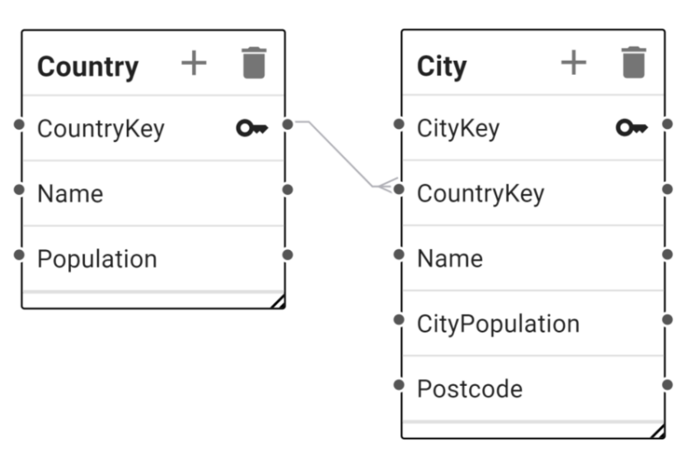
1 – 1, 1 – n, n –n Relationship

Primary key ref Primary key



1 – 1 relationship

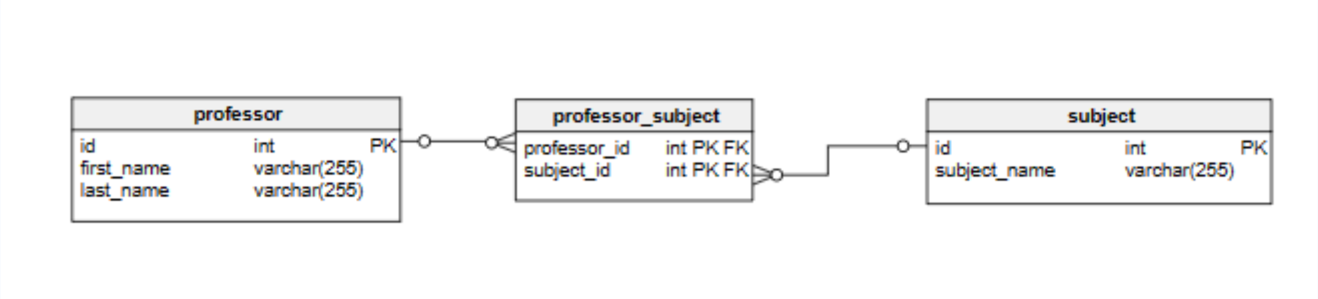
Foreign key ref Primary key



1 – n relationship

1 – 1, 1 – n, n –n Relationship

Use intermediate table



n – n relationship

Object Relational Mapping (ORM)

Instead of writing raw queries, we can map a **table** with a **model**

Most frameworks now support ORM, allow us to interact with database via Object

ORM provide simpler syntax, more readable than raw queries

Example: PHP Laravel

```
class User extends Model {

protected $table = 'my_users';

}
```

Laravel already cover all the setup we need in the **Model** class

Now we can interact with `my_users` table via **User** class

Eloquent ORM - Laravel 5.0 - The PHP Framework For Web Artisans

Example: PHP Laravel

```
$users = User::all(); // get all records
$user = User::find(1); //find record with primary key value is 1
$model = User::where('votes', '>', 100)->firstOrFail(); // complex queries
$user->save(); // update
```

Overview of Entity Framework Core - EF Core | Microsoft Learn

Sequelize | Feature-rich ORM for modern TypeScript & JavaScript

Eloquent ORM - Laravel 5.0 - The PHP Framework For Web Artisans

Non – Relational Database (NoSQL Database)

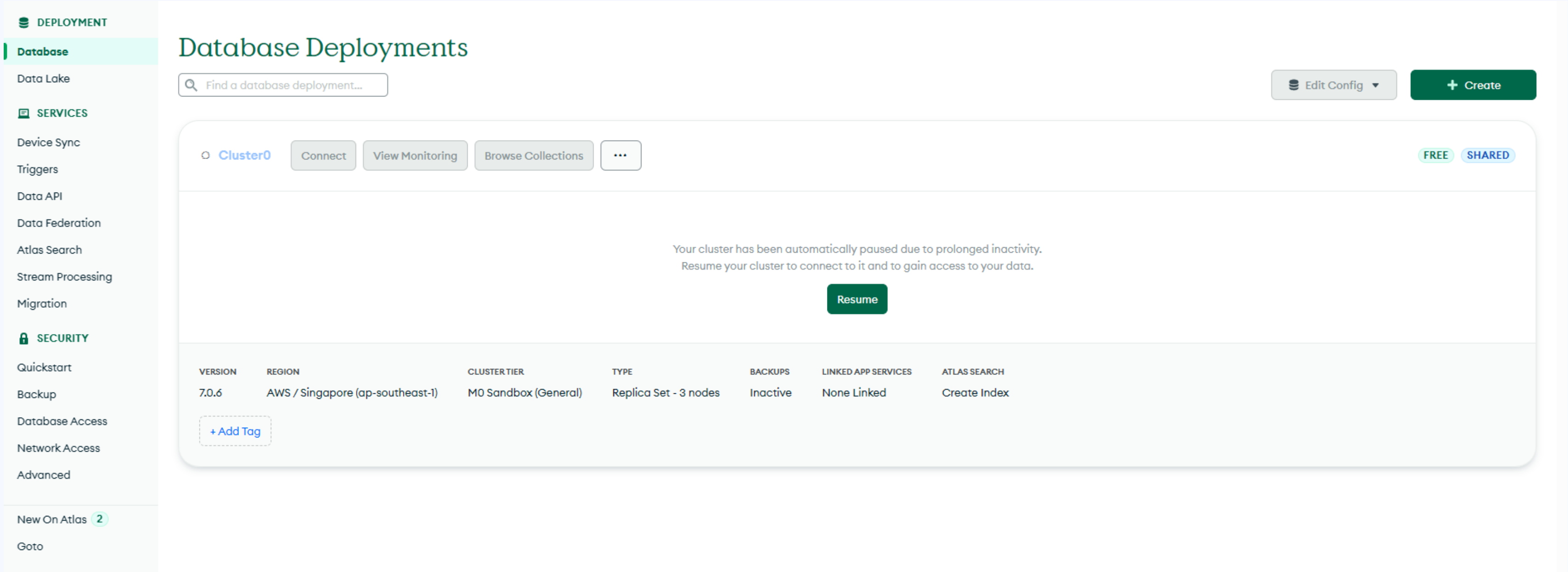
Storing data as **Collections** and **Documents** (just like **tables** and **records**)

More Flexible Schema than Relational Database

Syntax similar to ORM

MongoDB

Use Free Service to get start MongoDB Atlas | MongoDB



MongoDB

Get Connection string and connect with nodejs

Node.js MongoDB Create Database (w3schools.com)

Connecting with MongoDB Driver

1. Select your driver and version

We recommend installing and using the latest driver version.

Driver: Node.js

Version: 5.5 or later

2. Install your driver

Run the following on the command line

```
npm install mongodb
```

View MongoDB Node.js Driver installation instructions.

3. Add your connection string into your application code

View full code sample

```
mongodb+srv://fkmdev:<password>@cluster0.vnqljo6.mongodb.net/?
retryWrites=true&w=majority&appName=Cluster0
```

Firestore

Get started with Cloud Firestore | Firebase (google.com)

So sánh và Thao tác – MySQL và MongoDB

Yêu cầu:

- 1. Tạo bảng users trong MySQL (sử dụng PHPMyAdmin):
  - Cột id (INT, AUTO_INCREMENT, PRIMARY KEY)
  - Cột name (VARCHAR)
  - Cột email (VARCHAR)
- 2. Tạo collection users trong MongoDB Atlas:
  - Mỗi document có các trường name, email
- 3. Thêm ít nhất 3 bản ghi cho mỗi loại cơ sở dữ liệu với dữ liệu giống nhau.
- 4. Chụp ảnh màn hình kết quả (bảng hoặc document hiển thị).
- 5. So sánh và trả lời các câu hỏi sau (viết tay hoặc file txt):
  - Cấu trúc dữ liệu trong MySQL và MongoDB có gì giống và khác nhau?
  - Nếu bạn cần linh hoạt trong việc thay đổi cấu trúc dữ liệu, bạn chọn MySQL hay MongoDB? Vì sao?
  - Hãy nêu một ví dụ ứng dụng thực tế phù hợp với MongoDB hơn MySQL.